

High Input Impedance Chopper-stabilized Analog Front-end Based on Variable-capacitor Calibration for Bio-consumer Electronics

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Abstract—Analog front-end (AFE) circuits for bio-consumer electronics require ultra-high input impedance to ensure robust bio-potential acquisition under interference. This paper presents a chopping-stabilized AFE featuring an automatic input-impedance enhancement technique based on a positive feedback loop (PFL) with a continuously tunable variable-capacitor. An oscillation-detection-based calibration loop dynamically adjusts the PFL to compensate front-end parasitic capacitance, driving the system from an unstable condition to an optimal stable boundary with maximized input impedance while avoiding over-compensation. To improve calibration robustness, a frequency ripple suppression scheme is introduced, enabling precise control of the variable-capacitor independent of input frequency variations. The proposed variable-capacitance technique provides femtofarad-level resolution without relying on discrete capacitor arrays, achieving wide tuning range and high linearity. Fabricated in a 0.18 μm CMOS process, the AFE achieves an input impedance of 46.3 G Ω at 50 Hz, representing an 83 $\times$ improvement over conventional approaches. The prototype occupies 0.71 mm², achieves a mid-band gain of 40 dB, an input-referred noise of 0.93 μV_{rms} , and a power consumption of 3.48 μW . Experimental EEG and ECG measurements using comfortable dry electrodes validate the effectiveness of the proposed AFE for bio-consumer electronic applications.

Index Terms—Analog front-end, variable-capacitor, input impedance, positive feedback loop, bio-electronic devices.

I. INTRODUCTION

BIOELECTRIC signals, such as electrocardiograms (ECG), electroencephalograms (EEG), and electromyograms (EMG), provide real-time information about the physiological state of the human body [1]. These collected signals play a vital role in personalized telemedicine, intelligent health management, and unmanned control systems. Fig. 1 shows a remote healthcare scenario enabled by bio-consumer electronic devices, which can deliver personalized treatment tailored to individual patient characteristics while supporting long-term health management [2]. It incorporates a medical module for monitoring key physiological signals, such as ECG and EEG [3]. In addition, includes a battery, an analog front-end (AFE), and a data processing unit—comprising an analog-

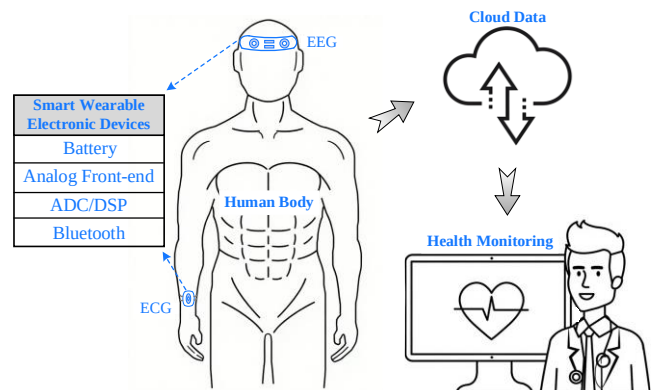


Fig. 1. Schematic diagram of consumer-grade smart electronic devices collecting bioelectric signals for health monitoring.

to-digital converter (ADC) and a digital signal processor (DSP)—as well as a transmission unit. Bioelectric signals are sensed through contact electrodes, transmitted by electrode leads to the AFE for amplification, and the acquired data are transmitted to the cloud or to a smart terminal for real-time visualization [4]. Obviously, the AFE, as a core component, determines the signal integrity of the entire readout system.

Numerous integrated-circuit (IC)-level techniques have been utilized to improve the signal quality of AFE for bio-potential acquisition [5]. For example, negative feedback architectures are widely employed to suppress interference, while differential-difference amplifiers and OTA stacking techniques have been explored to reduce input-referred noise. Additionally, common-mode duplication and differentially regulated enhancement schemes have been introduced to improve the common-mode rejection ratio (CMRR). However, many of these approaches rely on additional circuit blocks or auxiliary signal paths, which often increase design complexity, power consumption, and silicon area. Moreover, further solutions are required to address the issue of weak anti-interference capability caused by the AFE's low input impedance. This limitation mainly arises from the capacitive coupling between the contact electrode and the skin, which introduces a high-impedance interface at the AFE input. In practical bioelectric acquisition systems, especially wearable devices such as watches or smart glasses [6], the AFE is exposed to diverse electrode types and interface environments, resulting in significant input parasitic capacitances. These parasites originate from multiple sources, including the electrode-tissue interface, PCB routing, I/O pads, on-chip electrostatic discharge (ESD) structures, and amplifier input nodes. Within the frequency band of interest, the electrode-

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tissue interface exhibits a frequency-dependent impedance, and insufficient AFE input impedance leads to signal attenuation due to voltage division. The finite input impedance of the AFE can be equivalently characterized by an input capacitance at the operating frequency, which increases capacitive loading at the electrode interface and degrades signal integrity [7]. Although various circuit techniques have been proposed to enhance amplifier input impedance, suppressing the impact of aggregated parasitic capacitances and achieving sub-picofarad input equivalent capacitance remains challenging.

To address these challenges, an ultra-high input impedance AFE for bioelectric signal acquisition is proposed in this paper. The proposed AFE employs a positive feedback loop (PFL) with a variable-capacitor calibration (VCC) that automatically adjusts the strength of the positive feedback according to the oscillation amplitude. Moreover, the oscillation detection block regulates the AFE to search from unstable to steady state, maximizing the input impedance without stability problems. At the same time, the continuously tunable variable-capacitance with ultra-high resolution can accurately eliminate the parasitic capacitance of the input front-end. As a result, the proposed AFE achieves an input impedance of 46.3 G Ω at 50 Hz, with the input equivalent capacitance reduced to only 68 fF, while retaining an input-referred noise (IRN) of 0.93 μV_{rms} at 0.5–400 Hz.

The remainder of this paper is organized as follows. Section II reviews the state-of-the-art input impedance enhancement techniques. Section III describes in detail the proposed calibration mechanism for high input impedance and the circuit implementation of AFE. Section IV shows the proposed AFE acquisition prototype and the measurement results. Section V summarizes this work.

II. REVIEW OF HIGH INPUT IMPEDANCE TECHNIQUES

It is necessary to consider the construction of the input impedance interface in the AFE chip. Fig. 2 depicts a model of the electrode impedance interface with the distribution of the input parasitic capacitance. The commonly used electrode interfaces in acquisition applications are wet and dry electrodes [8]. Although wet electrodes have lower contact impedance, they typically have large dimensions (10 cm²) [9] and are unfriendly to users due to the need for gel. Dry electrodes have been used to solve this problem. In particular, small area (1 cm²) dry electrodes provide a better user experience than wet electrodes in terms of convenience and durability. However, the electrode-tissue impedance (ETI) of dry electrodes exhibits 10 – 20 times higher skin contact impedance and 1.5 times smaller capacitance than wet electrodes. This significantly attenuates the bioelectrical signals obtained by the AFE. Also, parasitic effects on the front end of the chip draw input current and attenuate useful information collected by the system. These parasitic effects include: the C_{PCB} for the adjacent metal traces on the PCB, the C_{PAD} for the chip's I/O pads, the C_{ESD} for the on-chip ESD, and the C_{INT} for the amplifier input node [10]. With the total input parasitic capacitance equal to 10 pF, the resulting total input impedance Z_{IN} is only 300 k Ω at 50 Hz, which imposes a large input equivalent capacitance on the AFE. The voltage division between the ETI and the input equivalent capacitance induces

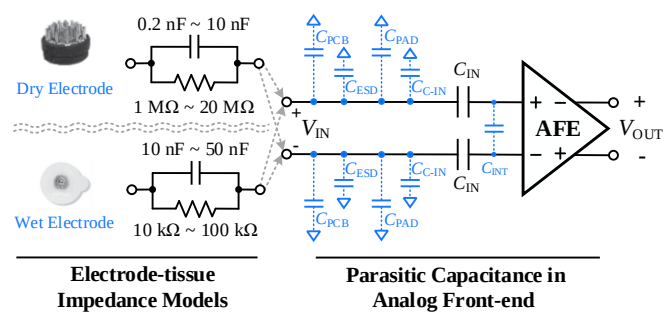


Fig. 2. Electrode impedance interface and input parasitic capacitance in AFE.

significant signal degradation and circuit instability [11].

A brief review of prior impedance-boosting techniques is presented to highlight the remaining challenges and motivate the proposed solution. Active shielding improves input impedance by bootstrapping parasitic capacitances [12], reducing signal-to-ground parasitics through shielding of signal lines. Careful layout, such as surrounding the V_{IN} PAD with the V_{OUT} PAD, is required to minimize unshielded C_{PAD} . The use of a local source follower further reduces input parasitic capacitance [13], achieving an input impedance of 6.7 G Ω at 50 Hz (equivalent to 0.47 pF). However, while external parasitics can be mitigated by driving the shielding layer, inherent parasitic capacitances remain and may compromise ESD robustness. Swapping the input and feedback paths corresponds to switching the input transconductance and represents a form of synchronous dynamic element matching (DEM) [14]. When the DEM and chopper frequencies are synchronized ($F_{\text{DEM}} = F_{\text{CH}}$), signal transfer occurs via charge redistribution rather than by charging input parasitics, significantly improving input impedance [15]. Input impedances exceeding 1 G Ω can be achieved; however, this technique is limited to current-feedback AFE architectures and does not support ultra-high impedances (>10 G Ω). Another approach involves pre-charging the input capacitors [16], which is insensitive to parasitic capacitance and effective at near-DC frequencies. An auxiliary buffer pre-charges C_{IN} before chopper operation, but since the auxiliary path is directly connected to the amplifier input [17], its DC offset and flicker noise are amplified, degrading AFE performance. Although this issue is partially alleviated in [18] by introducing an additional clock to drive a passive mixer, the buffer still consumes power and introduces significant noise. While an input impedance exceeding 1.6 G Ω is achieved from 0.1 to 20 Hz (equivalent to ~ 1 pF), such noise levels are suboptimal for bio-signal acquisition requiring noise below 5 μV_{rms} [19]. Moreover, since most bioelectric signals concentrate in the 10–50 Hz range [20], impedance enhancement in this band is of greater practical importance.

The above techniques are fundamentally constrained by the trade-off between low noise and high input impedance. The positive feedback loop (PFL) scheme effectively breaks this trade-off by using output feedback current to charge and discharge parasitic capacitances, preventing current draw from the signal source and thereby enhancing input impedance [21]. As the capacitive feedback redistributes current without introducing additional noise sources, PFL maintains low-noise

performance. The PFL is driven by the output of the amplifier and, according to Miller's theorem, the positive feedback capacitance C_{PF} can be viewed as a negative capacitance C_{NEG} [22] at the AFE's input node. C_{NEG} is expressed as follows:

$$C_{NEG} = (1 - A) \cdot C_{PF} \quad (1)$$

where A is the gain of the amplifier and it is much larger than 1. The approach of enhancing the input impedance of the AFE by introducing a C_{NEG} at the input node to counteract the parasitic capacitance has been proven to be effective [23]. Accurate capacitance matching enables high impedance over a wide range. In [24], a 4-bit fine impedance-boosting loop compensates for process variations. A coarse-fine tunable capacitor array is used in [25], achieving input impedance above 2 G Ω at 1 Hz after manual calibration. A three-terminal micro-adjustable capacitor achieves 1.5 G Ω from 0.1 to 10 Hz [26]. Proportional scaling of the input and feedback capacitors further increases impedance without degrading mid-band gain [27]. In [28], a local PFL decouples loop gain from AFE voltage gain, providing additional flexibility for parasitic compensation. An AFE based on a current-balancing architecture employs LPFL to cancel input transistor parasitics, achieving a minimum input impedance exceeding 1.8 G Ω at 5 Hz [29].

While PFLs provide an energy-efficient means to enhance the input impedance of analog front-ends, their effectiveness is highly sensitive to parasitic capacitance, process variation, and operating conditions, making calibration indispensable in practical implementations. Several calibration strategies have been reported to address this issue. An earlier approach in [29] adopts a digitally controlled impedance-boosting loop with coarse tuning granularity, prioritizing robustness at the expense of fine-resolution adjustment near the optimal operating point. In [19], a digitally assisted calibration scheme is employed to iteratively adjust the feedback strength and mitigate parasitic effects; however, the resolution is fundamentally limited by discrete tuning steps and requires careful stability margin design. The work in [5] introduces a background calibration approach that dynamically tracks impedance variations, but relies on digital control logic and periodic updating, which may increase system complexity and power consumption. More recently, [30] proposes a PWM-based autonomous calibration technique tailored to VCO-based neural front ends, where the feedback strength is modulated in the time domain; although effective, the calibration accuracy is inherently coupled to switching activity and frequency-dependent behavior.

These prior works demonstrate the PFL scheme is an effective means of enhancement and possesses natural advantages, yet there remains a need for improved resolution to more precisely compensate for parasitic capacitance. The necessity of calibration in PFL-based impedance enhancement, while also revealing trade-offs among tuning resolution, implementation complexity, and robustness. This issue has become a fundamental limiting factor in achieving input impedances exceeding 10 G Ω in existing PFL implementations, which is commonly regarded as a practical requirement in bioelectric acquisition systems. This work presents a solution to overcome these limitations and achieve ultra-high input impedance in PFL-based designs.

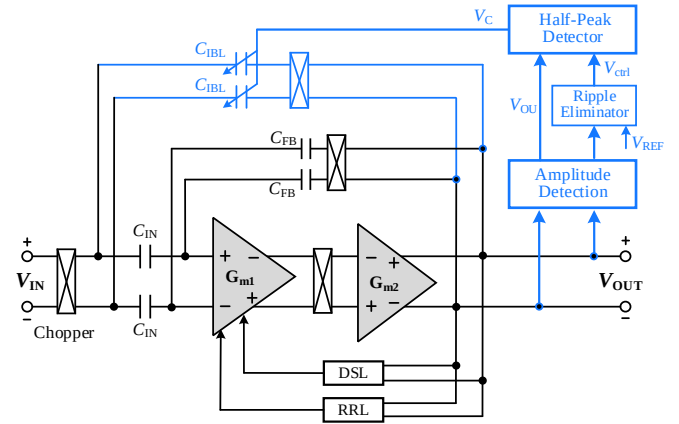


Fig. 3. Proposed overall system architecture for high input impedance AFEs.

III. IMPLEMENTATION OF THE PROPOSED ULTRA-HIGH INPUT IMPEDANCE AFE

A. Overall Structure

The proposed high impedance AFE, as shown in Fig. 3, comprises the following components: 1) a two-stage Miller-compensated OTA with chopper modulation, labeled as G_{m1} and G_{m2} ; 2) a DC servo loop (DSL) and a ripple reduction loop (RRL); 3) a PFL with a variable capacitor C_{IBL} ; 4) an amplitude detection block (amplitude detection, subtractor, and integrator), which is used to generate the control voltage V_C . The closed-loop gain of the AFE is determined by the ratio of the input capacitance C_{IN} to the feedback capacitance C_{FB} and is fixed at $C_{IN} / C_{FB} = 100$.

In the proposed AFE, the chopper structure allows the flicker noise of the two-stage operational amplifiers to be shifted from the baseband to high frequencies. Fig. 4 displays a schematic of a low-noise two-stage Miller-compensated OTA. Post-layout simulations indicate that achieves an open-loop DC gain of 84.6 dB with a phase margin of 50.5°, confirming adequate loop stability for the proposed AFE. The complementary input stage enhances the effective transconductance by combining contributions from both NMOS and PMOS differential pairs, thereby increasing overall gain without increasing static bias current. Under matched sizing and biasing, this structure reduces the IRN by approximately $\sqrt{2}$ compared to a single-polarity input stage [26], as the transconductance adds while the thermal noise sums in power. The large resistor R_F , connecting the input and output of G_{m1} , serves as a DC feedback loop and establishes a stable DC bias for the input differential pair. Notably, G_{m2} achieves high linearity with smaller input swing amplitude. As the noise of G_{m2} is much lower than that of G_{m1} , G_{m2} can be biased at a relatively lower current for better power efficiency. The implementation of Miller compensation ensures the stability of the closed loop. Additionally, common-mode feedback employs a pseudo-resistance sensing scheme to further enhance linearity.

Due to the chopper modulation, the resultant large electrode DC offset (EDO) can be coupled to G_{m1} through C_{IN} . The up-modulated EDO can saturate the AFE, resulting in the failure of the acquisition system. Typically, DSL technique mitigates EDO by extracting DC information at the output V_{OUT} . The AFE with DSL can suppress large electrode offsets that occur

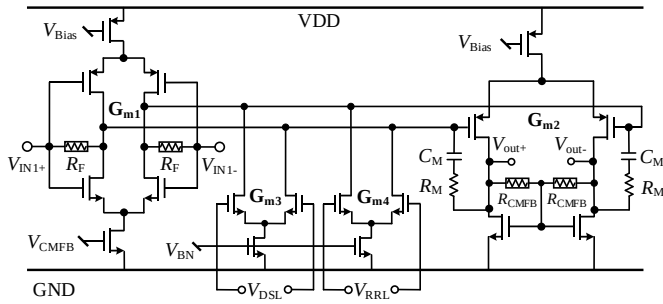


Fig. 4. Schematic of two-stage OTA with DSL and RRL transconductors.

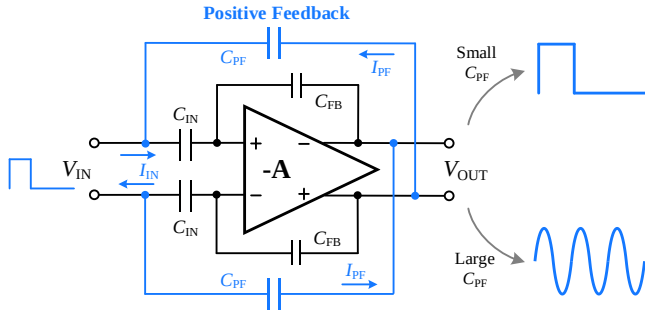


Fig. 5. AFE transient response of with a positive feedback loop.

during the acquisition process and can enable high-pass transmission of bioelectrical signals. A voltage-mode DSL is employed in the proposed AFE [23]. The switch-capacitor integrator-based DSL is used to attenuate low-frequency signals at the output V_{OUT} and suppress electrode offset voltages. The offset of G_{m1} is modulated by the frequency of the chopper, which can generate output ripple of up to several hundred millivolts. Such large output ripples may cause amplifier saturation and affect the performance of subsequent amplitude detection blocks. Therefore, an integrator-based RRL loop is employed in the feedback pathway [31]. This loop is enabled before impedance calibration, reducing the output ripple induced to sub-millivolt levels. Two NMOS input differential pairs labeled G_{m3} and G_{m4} are connected to the two-stage OTA to be used for the transconductance stages of DSL and RRL, respectively.

B. Principle of The Proposed Ultra-High Input Impedance

Adjusts the size of the positive feedback capacitor C_{IBL} based on the oscillatory conditions at the output end to achieve marginal stability of the AFE. This can almost completely compensate for the parasitic capacitance at the input end to achieve optimal input impedance. It is well known that the stability of PFL-based amplifiers can be evaluated by observing the output response to an input pulse [32]. Fig. 5 illustrates the AFE behavior under small-signal pulse excitation for different C_{PF} values. When C_{PF} is comparable to the feedback capacitance C_{FB} , the loop exhibits mild positive feedback and the AFE remains stable. In contrast, when C_{PF} significantly exceeds C_{FB} , the positive feedback dominates the negative feedback, resulting in ringing and oscillation. A single input pulse is sufficient to indicate whether the loop is over- or under-compensated, enabling fast and reliable calibration without continuous excitation. Accordingly, C_{PF}

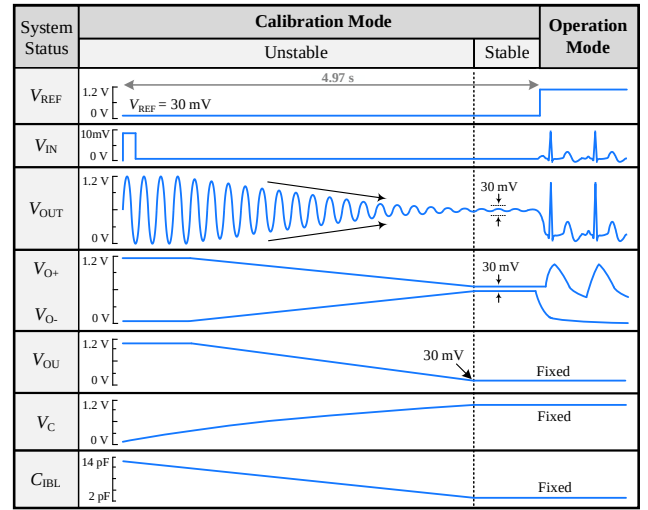


Fig. 6. Timing diagram of AFE in calibration and operation modes.

can be adjusted based on PFL stability criteria to achieve maximum parasitic capacitance compensation.

A VCC scheme with variable capacitors C_{IBL} has been integrated into the AFE to achieve ultra-high input impedance. In this system, the C_{IBL} serve the role of compensation and provide positive feedback gain, which is adjusted by a control voltage V_C produced by an amplitude detection block. This block consists of amplitude detectors, a subtractor, and an integrator. The operation involves the AFE's output positive and negative peaks being detected by the peak and valley detectors, respectively; the subtractor then calculates the amplitude of the output signal by subtracting the negative peak from the positive peak; the integrator compares this amplitude with a reference voltage V_{REF} , integrates their difference, and outputs the control voltage V_C . V_C adjusts the feedback strength of the PFL based on the output amplitude until the AFE transitions from oscillation to a stable state.

The proposed AFE features two modes: calibration mode and operating mode. The V_C is allowed to change only in the calibration mode, and the timing of its operation is depicted in Fig. 6. When rail-to-rail amplitude oscillations occur in the AFE output, the output V_{OU} of the subtractor continuously integrates, signaling to the system that the amplifier has saturated. At this point, the voltage V_C is monotonically increased, leading to an adjustment of C_{IBL} toward reducing the strength of positive feedback. As the auto-calibration proceeds, the positive feedback gain of the system is attenuated, and the amplitude of oscillation decreases. The C_{IBL} progressively approaches the parasitic capacitance at the front end of the AFE, thus gradually increasing the input impedance T_{ZIN} . The AFE moves from an oscillatory state to search for a stable boundary, with V_{REF} providing a reference for this boundary. Typically, an unstable state identified by oscillations require the amplitude of V_{OUT} to be approximately three times higher than the normal value [7]. In this work, a calibration input amplitude of 10 mV is selected, as it represents the upper bound of typical bioelectric signals and enables the calibration loop to converge under realistic operating conditions while maintaining sufficient voltage headroom. Accordingly, V_{REF} is set at a value of 30 mV. That is, when the oscillatory amplitude of the AFE output decreases

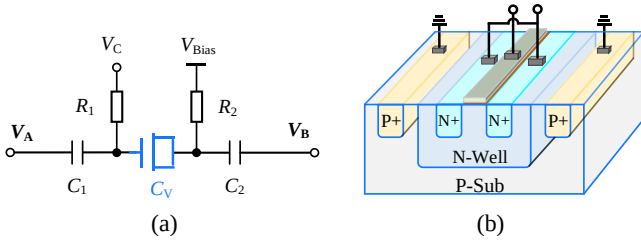


Fig. 7. (a) Circuit structure of variable-capacitor C_{IBL} , (b) 3-Dimensional structure of varactor C_V .

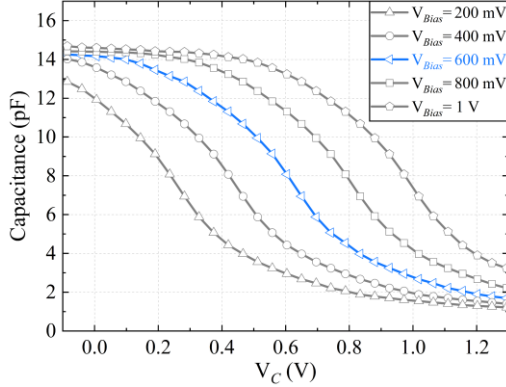


Fig. 8. Relationship between control voltage V_C of varactor C_V at different bias voltages.

to 30 mV, V_C no longer changes, providing a fixed capacitance C_{IBL} value. Auto-calibration ends when V_{REF} is high, and the AFE enters the operating mode with the highest input impedance. Thanks to the continuous adjustability of C_{IBL} rather than discrete adjustability, the calibration time of the AFE can be set as low as 5 seconds, still allowing enough margin before normal operation begins. This significantly shortens the calibration time compared to the calibration period of over 30 seconds reported in [21].

C. Structure of Variable Capacitors

A chopper-modulated variable-capacitor loop is connected in the positive feedback path between the input and output of the AFE, converting the output voltage to current which is then injected back into the input node. This current compensates for the current drawn by the chopper and input capacitor, as well as the current extracted by the parasitic capacitance from the signal source, thereby greatly increasing the input impedance of the AFE. The PFL topology based on continuously tunable variable-capacitors has three advantages. Firstly, voltage-controlled rather than switch-controlled variable capacitors provide extremely high capacitance resolution, breaking free from the limitations of traditional capacitor array LSBs. Secondly, voltage-controlled variable-capacitors do not introduce additional noise and gain mismatch, making them suitable for AFEs with various closed-loop gains. Thirdly, the capacitors with fixed layout structure connected to the inputs of the AFEs greatly avoid parasitic capacitance variations with the switching of capacitor arrays compared to conventional switched capacitor matrices.

The continuously adjustable C_{IBL} is depicted in Fig. 7, consisting of an accumulation-type MOS varactor diode C_V , MIM capacitors C_1 , C_2 , and resistors. V_A and V_B are the signal nodes, and V_C is the control node. Decoupling the signal nodes

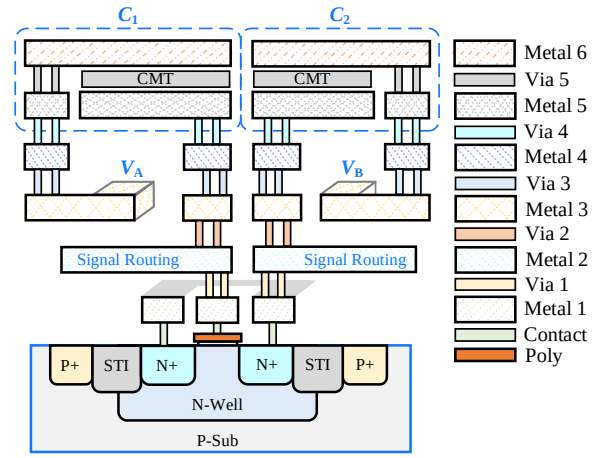


Fig. 9. Layout structure of variable-capacitor C_{IBL} .

TABLE I
DIMENSIONS OF THE PROPOSED VARIABLE-CAPACITOR

Capacitor	Size (μm)	Multiplier	Value (pF)
C_1	30×30	20	20
C_V	20×20	1	2–14
C_2	30×30	20	20

from the control node overcomes the limitation of traditional 2-T variable-capacitors whose capacitance values change with the output signal. The 3-D structure of C_V is shown in Fig. 7(b). The capacitance looking from the gate is simply the oxide capacitance. Therefore, there is a first-order relationship between the capacitance values C_V and V_{GS} in the near flat-band voltage:

$$C_V = \beta \cdot (V_C - V_{Bias}) \quad (2)$$

where β is a first-order coefficient. Under the control of gate voltage V_C , the oxide capacitance of C_V varies with the concentration of electrons and impurities. Fig. 8 shows the relationship between C_V and the V_C . The capacitance C_V is inversely related to V_C and directly related to the bias voltage V_{Bias} . The results demonstrate that the implemented C_{IBL} exhibits excellent controllability and good linearity over the operating range, as expected. By adjusting V_{Bias} , the C–V capacitance value can be shifted laterally, enabling flexible tuning under different operating conditions. The V_{Bias} is set to 0.6 V according to the AFE supply requirements, achieving a continuously adjustable capacitance exceeding 14 pF over a wide voltage range. Since C_V is embedded in a dynamically calibrated PFL, small voltage or temperature-induced variations are automatically compensated and therefore do not affect the steady-state gain or stability of the AFE.

Fig. 9 displays the layout of the implemented variable-capacitor. The C_V is positioned directly underneath the MIM capacitor to minimize routing length, and the critical interconnects are shielded by adjacent metal layers to reduce coupling and fringe parasitics. Table I summarizes the dimensions of the variable-capacitor components used in this work. The associated parasitic capacitance is static and absorbed into the calibrated total capacitance, and therefore

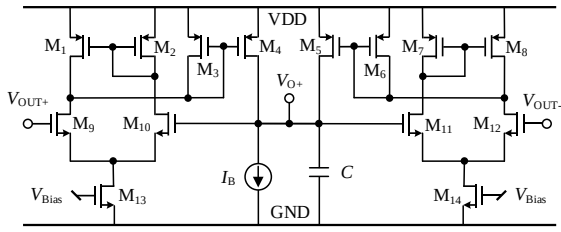


Fig. 10. Schematic diagram of peak detector.

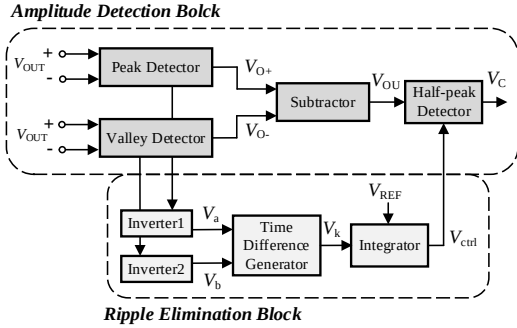


Fig. 11. Amplitude detection block with ripple elimination technique.

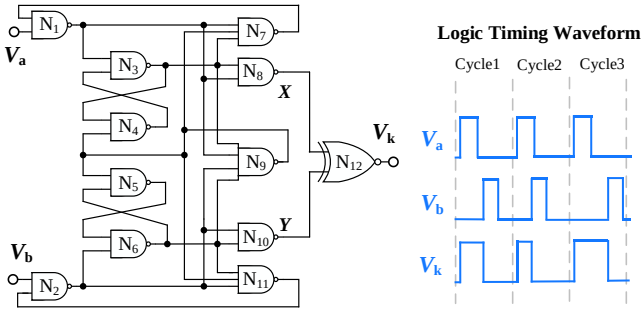


Fig. 12. Circuit and logic timing of the time difference generator.

does not affect the tunability, linearity, or high-impedance operation of the proposed AFE.

D. Amplitude Detection Block with Ripple Elimination

The V_C , which accurately reflects the oscillation amplitude of the AFE output, is critical for tuning the C_{IBL} . The peak and valley detectors jointly form an amplitude detection block that extracts the peak-to-peak oscillation by tracking the positive and negative envelopes of the output signal. The detected peak and valley voltages are further processed by a subtractor and a half-peak detector to generate a slowly varying control voltage V_C . As shown in Fig. 10, the peak detector adopts a differential-input, single-ended-output architecture, where two symmetrical 5-transistor OTAs detect the peaks of the in-phase (V_{OUT+}) and inverse-phase (V_{OUT-}) outputs, respectively. These detected peaks alternately charge the holding capacitor, ensuring that V_{O+} continuously tracks the maximum oscillation value. The valley detector operates in a complementary, inverted manner to track the minimum oscillation value. The subsequent half-peak detector is realized by the left-hand subcircuit in Fig. 10, which employs a 5-transistor OTA together with a current-mirror-capacitor network to achieve low-power peak tracking. A single-ended OTA compares the instantaneous input with the stored peak level and updates the

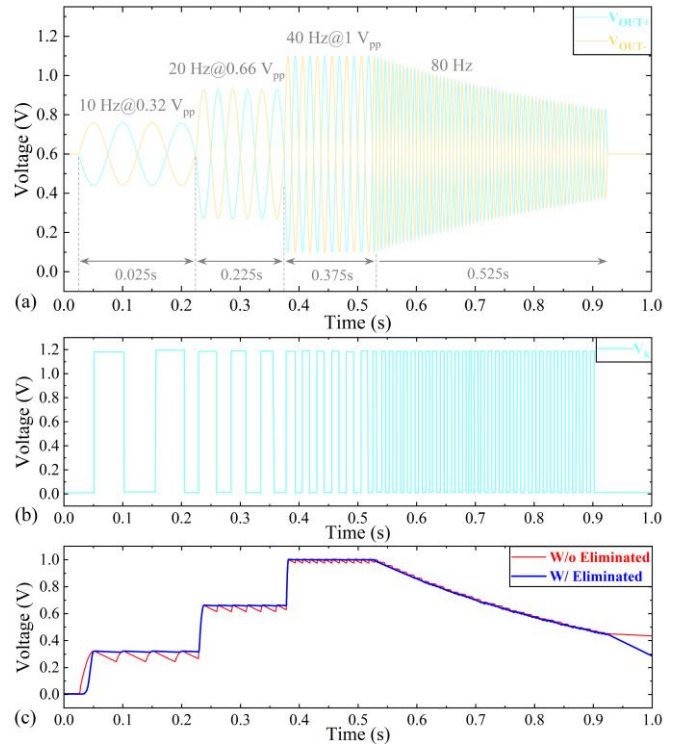


Fig. 13. The simulation results of ripple elimination, (a) differential output, (b) frequency-dependent pulse waveform output by the time difference generator, (c) comparison of frequency ripple elimination.

holding capacitor through a current-mirror charging path only when the input exceeds the stored value, while a controlled discharge current provides a slow reset, thereby generating a stable amplitude-representative control voltage.

Considering that the frequency of bioelectrical signals varies continuously in practical scenarios, this results in the inevitable production of ripple in the detected amplitude. This occurs because the charge on the capacitor C in Fig. 10 is constantly changing, and the current source I_B used for discharging cannot adjust at a corresponding rate. Additionally, the output voltage periodically decreases due to charge leakage. The ripple caused by the mismatched charging and discharging speeds affects the accuracy of the V_C .

This work implements a ripple elimination scheme within the amplitude detection block. As depicted in Fig. 11, this ripple elimination block comprises an inverter, a time difference generator [33], and an integrator. The inverter monitors the signal nodes in the peak and valley detectors that are correlated with the input frequency. The square wave information obtained is sent to the time difference generator (shown in Fig. 12) to produce a pulse waveform proportional to the input frequency. This pulse waveform's duty cycle serves as a control signal to adjust the charging and discharging speeds of the current source in the half-peak detector. In other words, the half-peak detector is controlled by the input signal frequency, allowing the charging and discharging speeds to automatically adjust in accordance with the signal frequency, unaffected by the signal amplitude.

The simulation results of the detection block under various frequencies and amplitudes are shown in Fig. 13. The input signals V_{IN+} and V_{IN-} are a set of differential signals, 0.32 V at

10 Hz during 0.025–0.225 s; 0.65 V at 20 Hz during 0.225–0.375 s; 1 V at 40 Hz during 0.375–0.525 s; subsequently, the amplitude gradually decreases at 80 Hz. It can be observed that the output curve without the ripple elimination technique is capable of detecting amplitudes, but exhibits ripple interference. Particularly at the low frequencies, there are significant ripples due to the mismatch of charging and discharging rates, and the superposition of these AC components causes the positive feedback gain to be unstable and the linearity of the variable capacitor to be deteriorated. In contrast, with the ripple elimination technique, ripples can be significantly mitigated and the amplitudes of the input differential signals can be clearly discerned even at varying frequencies. Comparative analysis of the two output voltage curves demonstrates that the proposed technique effectively eliminates ripples in the amplitude detection block, which is highly beneficial for the accurate extraction of AFE oscillation states.

Offset errors in the ripple elimination loop primarily introduce a small static shift in the V_C by slightly modifying the effective calibration decision threshold, without affecting the convergence behavior of the loop. Because the input impedance exhibits a relatively flat dependence on V_C around the optimum operating point, such offset-induced variations result only in second-order changes in input impedance and do not compromise system stability. Results under different threshold conditions further confirm the robustness of the ripple elimination loop against circuit offsets. In addition, the long-term stability of the proposed calibration mechanism is ensured through active maintenance of V_C . Rather than being passively stored on a capacitor, V_C is regulated by a low-bandwidth feedback path embedded in the half-peak detector, which dynamically compensates for leakage-induced drift and restores the equilibrium voltage. This slow regulation loop is effectively decoupled from the signal path and therefore neither interferes with normal operation nor induces oscillation, ensuring stable input impedance over time.

E. Noise Analysis

In order to analyze the noise of the proposed AFE, an equivalent noise model is developed as shown in Fig. 14. Note that thanks to the high open-loop gain of the two-stage op-amp, noise from the subsequent modules that are not in the PFL (e.g., the amplitude detection block and the ripple elimination block) is ignored. The power spectral density (PSD) of the IRN for AFE, $\overline{v_{n,CIA}^2}$, is calculated by dividing the output noise PSD by the mid-band gain, as shown below:

$$\overline{v_{n,CIA}^2} = \left[\overline{v_{n,OTA}^2} + \frac{g_{m3}^2}{g_{m1}^2} (\overline{v_{n,gm3}^2} + \overline{v_{n,RRL}^2}) + \frac{g_{m4}^2}{g_{m1}^2} (\overline{v_{n,gm4}^2} + \overline{v_{n,DSL}^2}) \right] \times \left(\frac{C_{IN} + C_{FB} + C_{DSL}}{C_{IN}} \frac{1}{1 + \frac{2\pi f_{hp}}{s}} \right)^2 \quad (3)$$

where g_{m1} , g_{m3} , and g_{m4} are the OTA transconductances as shown in Fig. 4; $\overline{v_{n,OTA}^2}$, $\overline{v_{n,gm3}^2}$, and $\overline{v_{n,gm4}^2}$ are the noise contributed by OTA and the transconductance stages of the

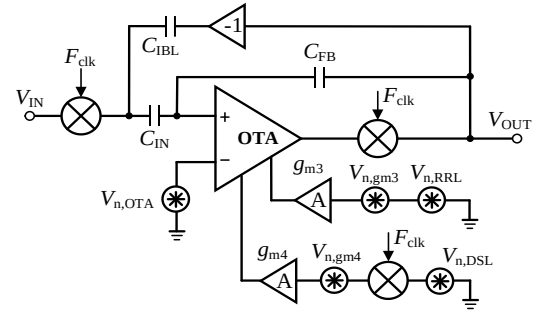


Fig. 14. Equivalent noise model of the proposed AFE.

RRL and the DSL, respectively; $\overline{v_{n,DSL}^2}$ and $\overline{v_{n,RRL}^2}$ are the output referred noise of the DSL and the RRL, respectively. For low noise designs, the input transistors in G_{m1} are biased in the deep subthreshold region. The current noise PSD can be modeled as (4).

$$\overline{i_{ni}^2} = 4kT \frac{1}{2s} g_m \quad (4)$$

where g_m is the transconductance of the input transistor and s is a process-dependent subthreshold slope factor. The subthreshold g_m is approximately equal to sI_D/V_T . The IRN PSD for a two-stage OTA can be expressed as (5).

$$\overline{v_{n,OTA}^2} = 4 \frac{\overline{i_{ni}^2}}{(2g_m)^2} = 2kT \frac{V_T}{s^2 I_D} \quad (5)$$

In (3), the noise from the DSL and RRL is not dominant because the flicker noise at the transconductance stage of the RRL and DSL is modulated to high frequencies by the chopper. The overall noise is attenuated by the large input transconductance G_{m1} of the OTA. For the RRL, the flicker noise of the OTA is also chopped out. Meanwhile, to minimize the current noise, 10 kHz chopping frequency, 32 pF input capacitor, and 0.5 / 0.18 μm NMOS chopping switches are used in this work.

IV. MEASUREMENT RESULTS

The proposed high input impedance AFE is fabricated using a standard 0.18 μm CMOS process, and the chip layout is shown in Fig. 15(a). The effective chip area is 0.71 mm^2 with the optimal matching using dummy technique. A detailed power analysis is shown in Fig. 15(b). The AFE chip consumes 2.9 μA at 1.2 V power supply, where the two-stage OTA dominates the power consumption due to its low noise requirement.

A. Input Impedance

The measurement setup is enclosed in a metal shielding box to suppress electromagnetic interference and ensure accuracy and repeatability. Before impedance measurement, the AFE is self-calibrated in its normal closed-loop configuration using a small-signal transient excitation to probe the stability of the positive feedback loop. The pulse response enables autonomous adjustment of C_{IBL} and convergence to the optimal high-impedance operating point without continuous excitation. After calibration, the control voltage V_C is fixed and the circuit enters operational mode for bioelectrical signal acquisition. The input impedance is then characterized using an indirect

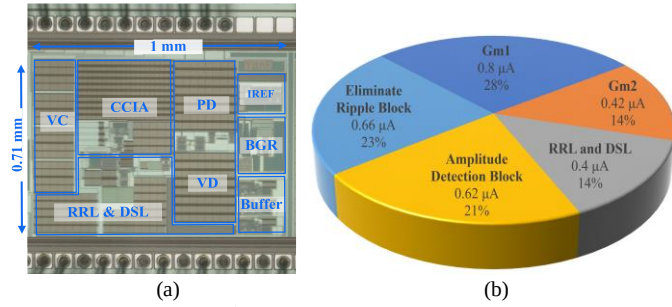


Fig. 15. (a) Micrograph of the high-input-impedance AFE chip. (b) Comparative analysis of power consumption between modules.

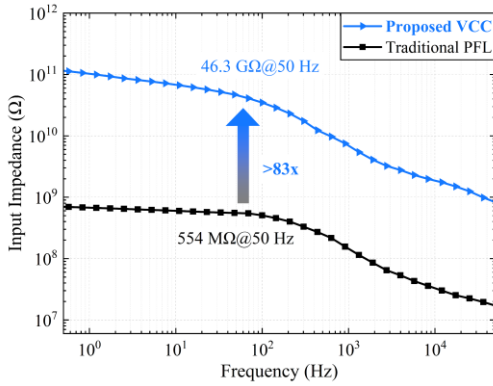


Fig. 16. Measurement results of input impedance versus frequency.

frequency-domain method by applying a small-signal AC excitation and extracting the impedance from the measured transfer response. Fig. 16 shows the test results of the relationship between input impedance and frequency, comparing the results obtained using the self-calibration technique with those from the traditional method. When implementing the conventional PFL, the measured input impedance at 50 Hz is 554 MΩ. After activating the VCC technique, the input impedance of the AFE reaches 46.3 GΩ at 50 Hz (an 83-fold increase), corresponding to an input equivalent capacitance of only 68 fF. Thanks to the automatic calibration of the PFL using the variable capacitor technology, the AFE can achieve significantly higher input impedance within the frequency range of interest than traditional techniques. Specifically, the proposed AFE achieves an input impedance of over 100 GΩ at 1 Hz. As a result, the input equivalent capacitance can be reduced to the femtofarad level, greatly minimizing the voltage division effect on the signal source.

To verify the high impedance capability of the AFE acquisition chip under different electrode contact environments, Fig. 17 compares the input impedance at 20 Hz with varying parasitic capacitance for both the traditional PFL method and the proposed variable capacitor self-calibration technique. In this test, the traditional PFL capacitor was set to 0.32 pF, equal to the feedback capacitance. As the assumed parasitic capacitance C_{P-EXT} increased from 1 pF to 9 pF, the traditional PFL, with its fixed capacitance value, was unable to adapt to the changing parasitic capacitance, causing the input impedance to drop from 478 MΩ to only 55 MΩ. However, the proposed VCC technique allows the feedback capacitance C_{IBL} to adaptively increase, providing more compensating current, thus maintaining an input impedance of 32 GΩ even when the external capacitance increases to 9 pF. At the same

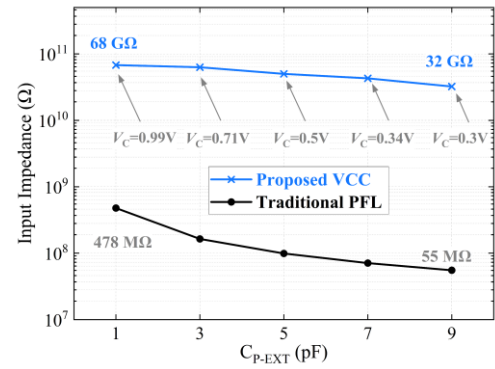


Fig. 17. Measured variation of T_{ZIN} at 20 Hz with various C_{P-EXT} s using the conventional PFL and the proposed VCC technique.

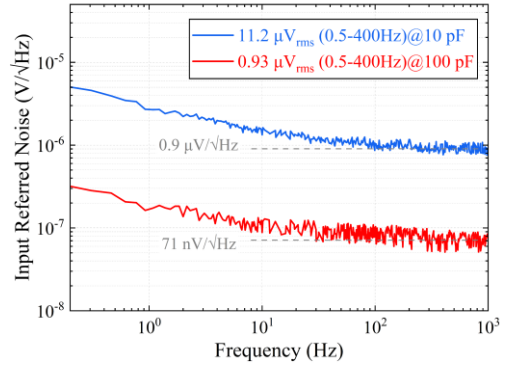


Fig. 18. Measured input referred noise power spectral density.

time, the control voltage V_C adjusts to changes in parasitic capacitance, modulating the strength of the positive feedback without any stability issues. This makes the proposed AFE an ideal choice for bioelectrical signal acquisition.

B. Characteristics of The Proposed AFE

The AFE exhibits a gain stability of 39.9 dB in the mid-frequency range, with a high-pass cutoff frequency of 0.35 Hz and a low-pass cutoff frequency of 5.4 kHz. This measurement characteristic is in excellent agreement with the simulation results obtained from periodic steady-state analysis, indicating its broadband performance that covers the full spectrum of bioelectrical signals. Fig. 18 presents the input reference noise measurement results of the proposed AFE. In the measurement, external capacitors of 10 pF and 100 pF were used to simulate the differences in input load caused by varying electrode contact tightness with the human body. Within the 0.5–400 Hz signal bandwidth, the measured integrated noise was 11.2 μV_{rms} and 0.93 μV_{rms}, respectively. Above 100 Hz, the corresponding circuit thermal noise densities were approximately 0.9 μV/√Hz and 71 nV/√Hz, respectively, demonstrating that the proposed AFE maintains good noise performance even under high input impedance requirements.

Fig. 19 presents the test results of the AFE's CMRR and power supply rejection ratio (PSRR) as a function of frequency under operating conditions. With a 0.6 V_{pp} common-mode input signal, the CMRR reaches 81 dB. The AFE's ultra-high input impedance effectively reduces the loading effect on the signal source, thereby minimizing common-mode signal variations and indirectly improving the CMRR. Furthermore, the introduction of chopper-stabilized further enhances the circuit's ability to suppress common-mode signals. Under a

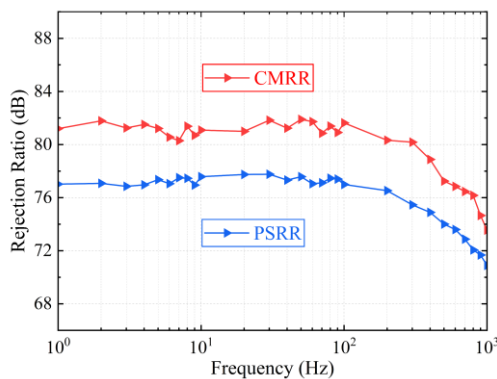


Fig. 19. Measurement results of CMRR and PSRR of the proposed AFE.

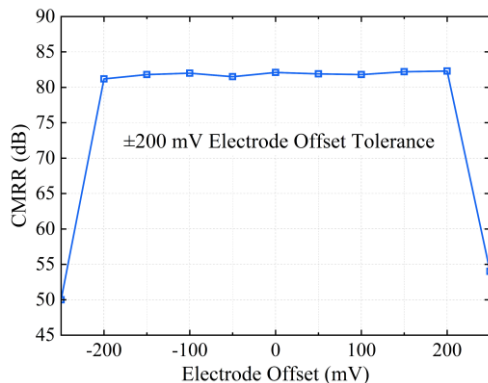


Fig. 20. Measured electrode offset versus CMRR.

0.1 V_{pp} power supply disturbance, the chip achieves a PSRR of approximately 77 dB. The higher PSRR indicates the AFE's strong ability to suppress power supply noise, which contributes to improved interference immunity of the acquisition system in complex power environments.

Fig. 20 shows the common-mode rejection characteristics of the AFE under different input offsets. The measurement results indicate that the AFE, equipped with VCC technique, effectively handles electrode offset, maintaining a CMRR of approximately 81 dB within the -200 mV to +200 mV range. This demonstrates its excellent interference rejection capability in high-impedance environments. When the input offset exceeds ± 200 mV, the CMRR decreases, with values of approximately 50 dB and 54 dB measured at -250 mV and +250 mV, respectively. These results suggest that the proposed AFE maintains stable common-mode suppression even in the presence of typical electrode offsets, effectively mitigating the impact of such offsets on the signal acquisition system's performance over a wide range. This ensures strong signal integrity for bioelectrical signal acquisition systems.

C. Biometric Measurement

The AFE acquisition test is primarily divided into two parts: ECG signals using dry electrodes; and scalp EEG signals using ear-clip electrodes. Fig. 21 show the testing environment for non-contact electrode-based ECG acquisition, where the dry electrodes have a 1 cm² contact area, and all electrodes are coupled to the skin through clothing. The acquired signals are transmitted via shielded coaxial cables to an oscilloscope, which is set to AC coupling mode, with a sampling rate of 20 kSa/s and a bandwidth of 500 Hz to ensure the integrity and preservation of the original signal characteristics during transmission.

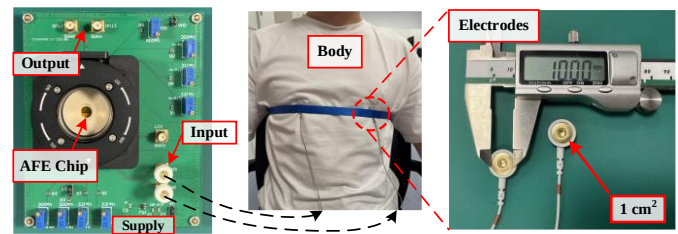


Fig. 21. The proposed AFE prototype uses non-contact electrodes for ECG signal acquisition.

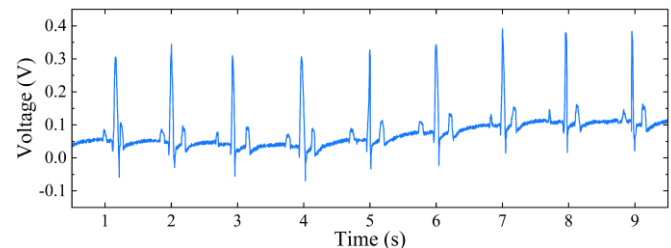


Fig. 22. ECG signal acquisition results.

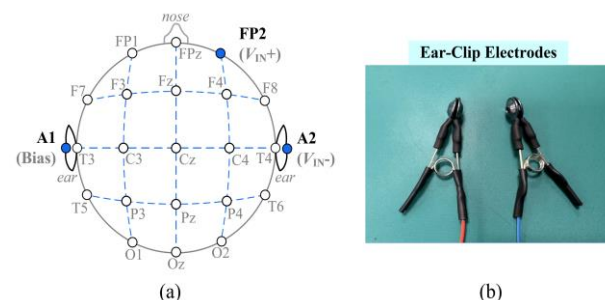


Fig. 23. (a) EEG signal acquisition configuration, (b) acquisition electrodes.

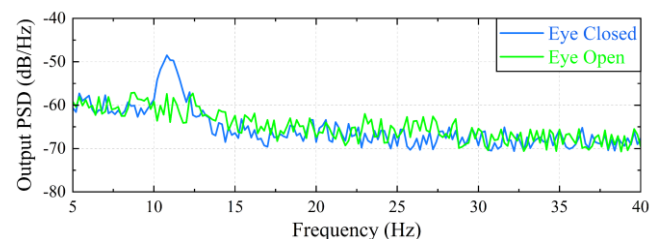


Fig. 24. EEG signal acquisition results.

Fig. 22 show the recorded ECG waveform. It is evident that the output waveform is not saturated, with clear peaks of the ECG and a well-defined QRS complex. The recorded waveform shows no significant power-line interference or myoelectric noise, in line with normal ECG characteristics. The observed baseline drift is mainly caused by the dry electrodes' contact through clothing, which is closely related to the tightness of the electrode fit. This type of drift is a controllable phenomenon and can be effectively mitigated by adjusting the electrode position or increasing the electrode contact, while still maintaining good wearing comfort and meeting practical application needs.

EEG signal acquisition configuration is shown in Fig. 23(a). Uses ear-clip electrodes to evaluate the adaptability of the AFE to different types of weak signals under various electrode contact impedances. The structure of the ear-clip electrode is shown in Fig. 23(b). It is comfortable to wear and allows for repeatable placement, effectively reducing electrode placement errors.

Fig. 24 presents the spectral analysis results of the acquired

EEG signals, where the approximately 11 Hz alpha wave activity in the closed-eye state is clearly identifiable, indicating that the system has strong signal feature resolution capabilities. The signal spectrum energy is primarily concentrated in the low-frequency range, with the overall noise floor well-controlled and excellent signal-to-noise ratio performance. This acquisition result further verifies that the proposed AFE design maintains excellent signal acquisition and amplification performance.

D. Performance Summary and Comparison

Table II summarizes the performance of the proposed AFE and compares it with other IAs used for biopotential acquisition. Some previous studies [32], [34] have focused on improving input impedance through active shielding techniques, which inevitably consume additional power and sacrifice noise efficiency. Although [21], [36] has demonstrated better noise efficiency at higher bandwidths, due to traditional PFL, their input equivalent capacitance $C_{in,eq}$ is very large. The proposed AFE, featuring an auto-calibrated PFL with variable-capacitance, achieves a high input impedance of 46.3 G Ω at 50 Hz, which is at the same level as existing advanced work. The proposed AFE has an input equivalent capacitance of 68.7 fF, which is worse than that of the reference [7], but has comparable noise efficiency and lower overall power consumption.

V. CONCLUSION

This paper proposes an AFE with high input impedance for bio-consumer electronics. The AFE, equipped with an oscillation detection circuit, actively adjusts the strength of the positive feedback based on the amplitude of oscillation to precisely compensate for the parasitic capacitance at the front end. The proposed ripple elimination block effectively suppresses the ripple, providing precise control voltage for the variable capacitor. The tunable variable-capacitor quickly finds stable boundaries in calibration mode without stability issues. The proposed AFE is designed using a standard 0.18 μ m CMOS process, consumes 3.48 μ W under a 1.2 V supply, and has an area of 0.71 mm². Thanks to the proposed variable-capacitor calibration technique, the AFE achieves an input impedance of 46.3 G Ω at 50 Hz. The AFE has an IRN of 0.93 μ V_{rms} within 0.5–400 Hz, while ensuring optimal input impedance and comparable noise performance. Validated through in-vivo ECG and EEG measurements, the proposed AFE is well-suited for bio-consumer electronics, enabling low-power, reliable signal acquisition for continuous health management.

TABLE II
PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART AFES

Parameter	JSSC [18] 2017	JSSC [32] 2018	VLSI [34] 2018	ISSCC [35] 2020	JSSC [21] 2022	JSSC [7] 2024	TVLSI [36] 2024	This Work
Process (μ m)	0.04	0.18	0.18	0.18	0.11	0.18	0.18	0.18
Supply (V)	1.2	0.8	5	1.8	1/1.5	1.2	1.8	1.2
Power (μ W)	2.8	0.29/ch	90	3.35	3.83	10.46	13.86	3.48
Gain (dB)	25.7	20-40	0/9/13/20	46-64	64/68/74	0	40	39.9
Area (mm ²)	0.069/ch	1.92	1.2	0.22	0.75	0.19	0.085	0.71
CMRR (dB)	85	66	N/A	102	N/A	95	120	81
IRN (μ V _{rms}) @ Hz	1.8 (1–200)	8.26 (1–400)	3.7 (0.5–100)	3.15 (0.5–5k)	0.36 (0.5–300)	0.72 (0.5–100)	0.59 (1–100)	0.93 (0.5–400)
Impedance (G Ω) @Hz	1@50	50@50	4@50	50@50	2@50	136@60	0.064@60	46.3@50
$C_{in,eq}$ (fF)*	3184	63	796	63	1592	19.5	41472	68.7
Self-Calibration	No	Yes	No	No	Yes	Yes	No	Yes
Boosting Techniques	Pre-charge	Active Shield & Self-calibration	Active Shield & Bootstrap	Active Shield & Trimming PFL	Dual PFL	Active Shield & C _{PF} Down Scaling	Small Input Capacitance	Variable-capacitor Calibration

* $C_{in,eq} = 1/2\pi T_{ZIN}F$.

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